Quantitative Branch and Corner Certificates for Near Inclusions of von Neumann Algebras

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Abstract

The assigned manuscript constructs one-sided near inclusions of von Neumann algebras with spatial embeddings but no implementing unitaries tending to the identity. We examine its actual operator-norm, Fourier, weak-limit, and finite-corner mechanisms. For the source clock/shift pair, centering the commutator path at its scalar value yields the finite separation $\|\widehat{U} - U_n\| + \|\widehat{V} - V_n\| \geq \cos(\pi/n)$. We quantify the corner lemma through the canonical projection-intertwining polar unitary and the subsequent polar corrections. The source Fourier majorant can be capped by unitarity, giving reordering constant 40768/3 instead of 206336/3; its discarded tails and matrix amplifications are controlled explicitly. Together these give an explicit, conservative implementing-unitary obstruction conditional on the source representation and whole-algebra hypotheses. A finite logarithm certificate retains the unnormalized trace, branch gap, and full physical matrix norm. Classical winding, polar decomposition, and Fourier estimates are credited; neither finite Gram data nor sampled generators certify the infinite-algebra construction. Contribution assessment: C2/5, explicit source-specific branch, corner, and Fourier budgets refining classical arguments without a new topological principle or independent global near-inclusion theorem.

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1 What the source asserts and what is checked

All operator norms are physical Hilbert-space operator norms. For unital von Neumann algebras $M, N \subseteq \mathcal{B}(H)$ with common identity, the source [1] uses

$$\gamma(M,N) = \sup_{x\in M,\,\|x\|\le 1} \inf_{y\in N} \|x-y\|\,. \tag{1}$$

The element y need not be in the unit ball. Define the one-sided unit-ball gap by the same formula with $\|y\| \le 1$. It lies between γ and 2γ: if $\|x-y\| < a$, then $\|y\| \le 1+a$, and replacing y by $y/\max(1,\|y\|)$ adds at most a. Take infima and then let a approach the relevant tolerance. The two-sided Kadison–Kastler distance is the maximum of the two unit-ball gaps. One-sided smallness supplies no reverse estimate.

If a unitary u satisfies $uMu^* \subseteq N$, then $\gamma(M,N) \le 2\,\|u-I\|$, by using uxu^* as approximant. The reverse implication is the disputed unrestricted assertion. The source claims separable examples (M_n, N_n) with $\gamma(M_n,N_n) \to 0$ and a uniform positive lower bound on every such implementing unitary's distance to I. It explicitly constructs an embedding, so this is not a non-embedding claim and not a two-sided counterexample.

1.1 Architecture of the seven cached texts

The main file, introduction, three proof modules, bibliography, and README have all been freshly read: 856 literal lines. The following map records the actual logical interfaces, not independent certification of the headline.

Module	Mechanism and retained requirement
`circle`	Tree vectors realize a unitary representation of the countably generated free group. A circle twist gives uniform Fourier majorants with a summable first moment and a specified weak limit.
`near`	Matrix substitution, a diagonal conjugating unitary, and an absolutely convergent four-index expansion give an approximant on the entire source algebra, not just its generators.
`obstruction`	The trace-log invariant separates clock/shift from commuting pairs. Weak cluster points in commuting algebras produce nearby support projections, a common finite corner, and commuting polar parts.

The source uses no product-of-weak-limits inference. Its commuting limit algebras come from an actual representation of the product group. Likewise, it does not assume arbitrary compression preserves commutation. These are valid structural details that must survive any quantitative refinement.

1.2 The original representation data

Let F be the free group on generators s_j, $K = \ell^2(F)$, and p_e the projection onto the identity vector. The source fixes $t = 1/2$, $r = \sqrt{1-t^2}$, and defines along the rooted Cayley tree $\xi_e = \delta_e$, $\xi_x = t\xi_{p(x)} + r\delta_x$. Their Gram entries are $t^{d(x,y)}$, by separating the two orthogonal tails after their last common ancestor. The span contains every basis vector, and left translation preserves these Gram entries; hence $\pi(g)\xi_x = \xi_{gx}$ extends to a unitary representation.

If $T_g = \pi(g)l(g)^*$, the source computes its difference from the identity on the path $e = h_0, \dots, h_d = g$. On that path,

$$T_g\delta_{h_b} = r\xi_{h_b} - t\delta_{h_{b+1}}\ (b<d), \qquad T_g\delta_{h_d} = \xi_{h_d}, \qquad |(T_g)_{ab}| \le t^{|a-b|}.$$

With the character $q(s_j) = 1$ and diagonal $D_z\delta_x = z^{-q(x)}\delta_x$, put $\pi_z(g) = z^{q(g)}D_z\pi(g)D_z^*$. It has

$$\pi_z(g) = \sum_{k\in\mathbb{Z}} z^k A_k(g), \qquad \sup_g \|A_k(g)\| \le c_k := \mathbf{1}_{k=0} + 4\,2^{-|k|}. \tag{2}$$

The path row and column sums bound each Fourier coefficient, because an exponent k requires path distance at least $|k|$. The source also obtains $\pi_z(s_j) \to tzp_e$ weakly. Uniform norm bounds and basis-vector convergence justify this passage. The finite path computation is not a claim that a finite Gram matrix controls the whole Hilbert space; the dense-span extension and uniform estimates do.

2 The finite topological obstruction with its branch margin

For unitaries $U, V \in M_n(\mathbb{C})$, write $W = UVU^*V^*$. If $-1 \notin \operatorname{spec} W$, the principal logarithm is defined and

$$\kappa(U,V) = \frac{1}{2\pi i}\operatorname{Tr}\log W \in \mathbb{Z}. \tag{3}$$

Here Tr is unnormalized. The eigenangles of W lie in $(-\pi,\pi)$; their sum is an integer multiple of 2π, since $\det W = 1$. Functional calculus makes this integer continuous, hence constant, along every unitary-pair path avoiding the cut. This is the classical winding obstruction of Voiculescu and Exel–Loring, in trace-log form [2]. Our orientation is UVU^*V^*; reversing it reverses the sign. A normalized trace would give κ/n, not an integer.

Proposition 2.1 (A spectral-gap path certificate)**.** *Suppose $g = \operatorname{dist}(-1, \operatorname{spec}(UVU^*V^*)) > 0$. Let $\widehat{U}, \widehat{V}$ be unitaries with errors $a = \left\|\widehat{U} - U\right\| < 2$, $b = \left\|\widehat{V} - V\right\| < 2$. If $2(a+b) < g$, their trace-log integers agree. In particular, for the clock/shift pair*

$$U_n e_j = \omega_n^j e_j, \qquad V_n e_j = e_{j+1 \bmod n}, \qquad \omega_n = e^{2\pi i/n}, \quad n \ge 3, \tag{4}$$

every commuting unitary pair satisfies

$$\left\|\widehat{U} - U_n\right\| + \left\|\widehat{V} - V_n\right\| \ge \cos(\pi/n). \tag{5}$$

Proof. The relative unitaries have principal self-adjoint logarithms: $U^*\widehat{U} = e^{iA}$, $V^*\widehat{V} = e^{iB}$, with spectra in $(-\pi,\pi)$. Along $U_s = Ue^{isA}$, $V_s = Ve^{isB}$, $0 \le s \le 1$, their deviations from U, V are at most a, b, since $|e^{is\theta} - 1| \le |e^{i\theta} - 1|$. Four-factor telescoping gives

$$\|U_sV_sU_s^*V_s^* - UVU^*V^*\| \le 2(a+b).$$

Normality gives $\|(W+I)^{-1}\| = 1/g$. The perturbation bound and Neumann's criterion keep $U_sV_sU_s^*V_s^* + I$ invertible. The path therefore avoids -1 and the integer is unchanged.

For (4), $U_nV_n = \omega_n V_nU_n$, so $W = \omega_n I$, $\kappa = 1$, and $g = |1 + \omega_n| = 2\cos(\pi/n)$. If the sum of errors were smaller than $\cos(\pi/n)$, both errors would automatically be smaller than two, and the first part would give $1 = 0$, since a commuting pair has $W = I$. □

The source centers its four-factor path estimate at I, obtaining the sufficient guard $\nu_n + 2(a+b) < 2$, where $\nu_n = 2\sin(\pi/n)$. Centering at the actual scalar commutator retains the larger gap $2\cos(\pi/n)$. Both are valid. The resulting bound is not asserted to be the optimal distance to commuting pairs. No continuous eigenvalue labeling is needed.

2.1 A certifiable logarithm, not a branch chosen by plotting

If $\rho = \|W - I\| < 1$, define $L_J(W) = \sum_{j=1}^{J} (-1)^{j+1}(W-I)^j/j$. Absolute operator-norm convergence gives

$$\|\log W - L_J(W)\| \le \frac{\rho^{J+1}}{(J+1)(1-\rho)}, \qquad \left|\kappa - \frac{\operatorname{Im}\operatorname{Tr} L_J(W)}{2\pi}\right| \le \frac{n\rho^{J+1}}{2\pi(J+1)(1-\rho)}. \tag{6}$$

Indeed bound each remaining $1/j$ by $1/(J+1)$, sum the geometric tail, and use $|\operatorname{Tr} A| \le n\,\|A\|$. A further certified trace-evaluation error is added to this scalar budget. If the resulting interval contains exactly one integer, that integer is κ. Merely obtaining a decimal close to one is not the same certificate. The factor n must not be discarded as if the trace were normalized. For clock/shift, $\rho < 1$ holds when $n \ge 7$; smaller dimensions still satisfy the branch proof above, but this particular series bound is not available.

The fresh original Ram analytic-continuation theorem [5] requires a common branch/domain and locally uniform majorants, and expressly distinguishes uniqueness from stability. Formula (6) provides such a concrete majorant only on its stated operator-norm ball. Neither continuation from finitely many samples nor a finite jet can legitimize crossing the branch cut.

3 Quantitative recovery of the common finite corner

Let E_1, E_2 be commuting orthogonal projections on a complex Hilbert space H, with $E = E_1E_2$ having nonzero finite-dimensional range D. Let $\widetilde{R}_i$ be unitaries commuting with both projections, and write $R_i = \widetilde{R}_i|_D$. The R_i need not commute. Suppose $\mathcal{A}_1, \mathcal{A}_2 \subseteq \mathcal{B}(H)$ are commuting unital von Neumann algebras and $T_i \in \mathcal{A}_i$ satisfy

$$\left\| T_i - E_i\widetilde{R}_i \right\| \le \delta_i \quad (i = 1, 2). \tag{7}$$

All hypotheses concern the full operators, including components outside D, not their compressions alone.

Lemma 3.1 (Canonical projection intertwiner)**.** *For projections E, P with $\|P - E\| = e < 1$, put $S = PE + (I-P)(I-E)$, and let $J = S(S^*S)^{-1/2}$. Then*

$$JEJ^* = P, \qquad \|J - I\| = j(e) := \sqrt{2 - 2\sqrt{1-e^2}} \le \sqrt{2}\,e. \tag{8}$$

Proof. Direct multiplication gives

$$S^*S = SS^* = I - (P-E)^2, \qquad (S + S^*)/2 = S^*S.$$

This positive operator is bounded below by $(1-e^2)I$, so S is invertible and normal. Also $SE = PS$, and S^*S commutes with E; hence $JE = PJ$. Normality implies S commutes with its absolute value and $\operatorname{Re} J = |S| = \sqrt{I - (P-E)^2}$. Consequently $(J-I)^*(J-I) = 2I - 2|S|$. The spectral theorem gives the equality in (8); rationalizing yields $j(e)^2 = 2e^2/(1 + \sqrt{1-e^2}) \le 2e^2$. □

Theorem 3.2 (Finite corner error budget)**.** *Under* (7)*, define*

$$d_i = \delta_i(2 + \delta_i), \qquad e_* = 2(d_1 + d_2), \qquad c_i = \delta_i + 2j(e_*).$$

Assume $d_i < 1/2$, $e_* < 1$, and $c_i < 1$. Then there are commuting unitaries $\widehat{R}_1, \widehat{R}_2$ on D with

$$\left\| \widehat{R}_i - R_i \right\| \le 2c_i, \qquad \sum_i \left\| \widehat{R}_i - R_i \right\| \le 2(\delta_1 + \delta_2) + 8j(e_*). \tag{9}$$

Proof. Since $(E_i\widetilde{R}_i)^*(E_i\widetilde{R}_i) = E_i$, $\|T_i^*T_i - E_i\| \le d_i$. Spectral enclosure about the projection puts $\operatorname{spec}(T_i^*T_i)$ in $[0, d_i] \cup [1 - d_i, 1 + d_i]$. Define $Q_i = \mathbf{1}_{(1/2,\infty)}(T_i^*T_i) \in \mathcal{A}_i$. Functional calculus gives $\|Q_i - T_i^*T_i\| \le d_i$, and hence $\|Q_i - E_i\| \le 2d_i$. Because the two algebras commute, $P = Q_1Q_2$ is a projection with $\|P - E\| \le 2(d_1 + d_2) = e_*$. The preceding lemma supplies a unitary J with $JEJ^* = P$ and $\|J - I\| \le j(e_*)$. In particular PH and D have the same finite dimension.

The operators PT_1P, PT_2P commute across adjoints. To verify this instead of assuming it, let $A = Q_1T_1Q_1 \in \mathcal{A}_1$ and $B = Q_2T_2Q_2 \in \mathcal{A}_2$. Then the compressions are AQ_2 and Q_1B; both products are AB. Replacing either factor by its adjoint proves the remaining commutations. Transfer to D by $C_i = (EJ^*T_iJE)|_D$. Comparison with the norm-one partial unitary $E_i\widetilde{R}_i$ before conjugation gives

$$\|C_i - R_i\| \le \delta_i + 2\,\|J - I\| \le c_i < 1.$$

Thus C_i is invertible on D. Its singular values lie in $[1 - c_i, 1 + c_i]$, since R_i is unitary. For its polar unitary $\widehat{R}_i = C_i(C_i^*C_i)^{-1/2}$, this implies $\left\|\widehat{R}_i - C_i\right\| = \|I - |C_i|\| \le c_i$. The unital C^*-algebras generated by C_1, C_2 commute, so their polar unitaries commute as well. The triangle inequality completes (9). □

This proof quantifies the source's existing corner argument; it does not replace it by an assertion about a finite Gram matrix. The spectral threshold, common range, intertwiner, and polar parts all have separate margins. If any displayed premise fails, the certificate is inconclusive rather than a proof that no commuting approximation exists.

Corollary 3.3 (Clock/shift corner obstruction)**.** *Suppose R_1, R_2 on $D = \mathbb{C}^n$ are the clock/shift pair. Data* (7) *satisfying the theorem's hypotheses cannot also satisfy*

$$2(\delta_1 + \delta_2) + 8j\big(2(d_1 + d_2)\big) < \cos(\pi/n). \tag{10}$$

In particular, common errors $\delta_i \le 1/250$ are impossible for every $n \ge 3$.

Proof. The first assertion combines (5) and (9). For $\delta = 1/250$, $e_* = 4\delta(2 + \delta) = 501/15625$. Using $j(e_*) \le \sqrt{2}e_* < 3e_*/2$, the left side is at most

$$4\delta + 12e_* = \frac{6262}{15625} < \frac{1}{2} \le \cos(\pi/n).$$

The same estimates give $d_i < 1/2$, $e_* < 1$, and $c_i \le \delta + 3e_* < 1$. Monotonicity covers smaller errors. □

4 A capped Fourier majorant on the whole algebra

The source bound $c_k = \mathbf{1}_{\{k=0\}} + 4\,2^{-|k|}$ is not the only available bound. With normalized Haar measure on the unit circle,

$$A_k(g) = \int z^{-k}\pi_z(g)\,dz, \qquad \|A_k(g)\| \le 1,$$

because the integrand is unitary. This is the classical Fourier coefficient bound, applied before summing the source's reordering estimate. Thus use

$$d_k = \min(1, c_k) = \begin{cases} 1, & |k| \le 2, \\ 4\,2^{-|k|}, & |k| \ge 3. \end{cases} \qquad S_0 = \sum_k d_k = 7, \quad S_1 = \sum_k |k| d_k = 14. \tag{11}$$

The dependence on the actual unitary family matters: a general operator-valued sequence satisfying the old majorant need not satisfy this cap.

Proposition 4.1 (Exact majorant constant)**.** *For the source Fourier construction, the four-index reordering constant can be taken to be*

$$C_d = S_0 S_1 \sum_{m,p\in\mathbb{Z}} d_m d_p |m-p| = \frac{40768}{3}. \tag{12}$$

For comparison, evaluating the same expression with the original c_k gives $206336/3$*, whereas its elementary product-moment bound is* 86528*. These are upper certificates from majorants, not evaluations of the actual algebra gap.*

Proof. For any nonnegative symmetric summable sequence w, of total mass S and finite first moment, let $A_j = \sum_{k>j} w_k$. Each distance $|m-p|$ counts the integer cuts separating m, p. Tonelli therefore gives

$$\sum_{m,p} w_m w_p |m-p| = 2\sum_{j\in\mathbb{Z}} A_j(S-A_j) = 4\sum_{j\geq 0} A_j(S-A_j).$$

Symmetry in the last equality pairs the cut at j with that at $-j-1$. For $w = d$, the tails are $A_0 = 3, A_1 = 2$, and $A_j = 4\,2^{-j}$ for $j \geq 2$. Hence

$$\sum_{m,p} d_m d_p |m-p| = 4\left(12 + 10 + 7\sum_{j\geq 2} 4\,2^{-j} - \sum_{j\geq 2} 16\,4^{-j}\right) = 4(12+10+14-4/3) = 416/3.$$

Multiplication by $7\cdot 14$ proves (12). For c, $S = 13, S_1 = 16, A_j = 4\,2^{-j}$ for $j \geq 0$; the distance sum is $4(104 - 64/3) = 992/3$. The bound $|m-p| \leq |m| + |p|$ gives $2S_0^2 S_1^2 = 86528$ for that original sequence. □

4.1 Why this estimates the entire unit ball

Here are the algebraic and analytic details retained from the source. Put $G = F \times F$, $\mathcal{L} = K \otimes K \otimes D$, and

$$a(g) = \sum_k A_k(g) \otimes I \otimes U^k, \qquad b(g) = \sum_k I \otimes A_k(g) \otimes V^k.$$

Finite spectral decomposition of U, V makes each a unitary representation; their ranges need not commute. Define $f(x) = a(x_1)b(x_2)$. On $H = \ell^2(G) \otimes \mathcal{L}$, let W_f be the diagonal unitary with entries $f(x)$, and set

$$M = W_f(\lambda(G)'' \otimes I)W_f^*, \qquad N = (\rho(G) \otimes I)'.$$

For $X \in \lambda(G)''$, put $Z_X = W_f(X \otimes I)W_f^*$. Its matrix blocks are

$$(Z_X)_{xy} = X_{xy} a(x_1)b(x_2)b(y_2)^* a(y_1)^*.$$

The reordered candidate has blocks

$$(Y_X)_{xy} = X_{xy} a(x_1)a(y_1)^* b(x_2)b(y_2)^* = X_{xy} f(xy^{-1}). \tag{13}$$

It is essential first to prove boundedness; formal block entries alone do not define a bounded operator.

Let D_{km} be diagonal on $\ell^2(G) \otimes K \otimes K$, with entry $A_k(x_1) \otimes A_m(x_2)$. Then $\|D_{km}\| \leq d_k d_m$, and, writing $B_{kmph}(X) = D_{km}(X \otimes I \otimes I)D_{hp}^*$, we have

$$Z_X = \sum_{k,m,p,h} B_{kmph}(X) \otimes U^k V^{m-p} U^{-h}, \tag{14}$$

$$Y_X = \sum_{k,m,p,h} B_{kmph}(X) \otimes U^{k-h} V^{m-p}. \tag{15}$$

Both series converge absolutely in operator norm, bounded by $S_0^4 \|X\|$. Right simultaneous translation preserves (13), since $X_{xq,yq} = X_{xy}$. Thus this bounded Y_X belongs to N. Integer telescoping, including inverse powers, gives

$$\left\|U^a V^b - V^b U^a\right\| \le |a||b|\,\nu, \qquad \nu = \|UV - VU\|\,.$$

Subtracting the two convergent series proves

$$\|Z_X - Y_X\| \le C_d \nu \,\|X\|\,, \qquad \gamma(M,N) \le C_d \nu. \tag{16}$$

No passage from bounds on a list of generators to bounds on arbitrary polynomials in those generators has been used.

Proposition 4.2 (An actual completely bounded comparison map)**.** *The map $\Theta : M \to N$, $\Theta(Z_X) = Y_X$, is unital and linear, and*

$$\|\Theta - \iota\|_{\mathrm{cb}} \le C_d \nu, \qquad \|\Theta\|_{\mathrm{cb}} \le 1 + C_d \nu,$$

where $\iota : M \hookrightarrow \mathcal{B}(H)$ is inclusion. This is a property of the displayed construction, not an implication from one-sided closeness for arbitrary von Neumann algebras.

Proof. Uniqueness of X makes the definition meaningful; (13) shows linearity and $Y_I = I$. At matrix level q, use $I_q \otimes D_{km}$ and $X \in M_q(\lambda(G)'')$ in the same absolutely convergent series. Every majorant and commutator bound is independent of q. Taking the supremum over q proves the claim. □

No multiplicativity or positivity is asserted. For group elements from opposite factors, the multiplicative defect contains $[b(s), a(t)]$. A completely bounded near-identity linear map must not be silently replaced by a spatial homomorphism.

4.2 Certified tails and what a finite truncation loses

For an integer $R \ge 2$, define $S_{j,R} = \sum_{|k|\le R} |k|^j d_k$ (with $j = 0$ interpreted without 0^0). Direct geometric summation gives

$$S_{0,R} = 7 - 8\,2^{-R}, \qquad S_{1,R} = 14 - 8(R+2)\,2^{-R}. \tag{17}$$

Retaining the cube $|k|, |m|, |p|, |h| \le R$ in the reordering difference costs at most $C_R \nu \,\|X\|$, where

$$C_R = S_{0,R} S_{1,R} \sum_{|m|,|p|\le R} d_m d_p |m - p|.$$

On its complement, use $|m - p| \le |m| + |p|$ only there. The discarded difference is bounded by

$$2\big(S_0^2 S_1^2 - S_{0,R}^2 S_{1,R}^2\big)\nu \,\|X\|\,. \tag{18}$$

Indeed the total sum of the nonnegative product-moment majorant is $2S_0^2 S_1^2$, and its sum on the cube is $2S_{0,R}^2 S_{1,R}^2$. Each individual operator series has discarded norm at most $(S_0^4 - S_{0,R}^4)\,\|X\|$. These bounds are uniform at all matrix levels. Importantly, a truncated Y_X need not commute with right translations: the representation identity used in (13) is a full-series identity. Truncations certify errors to the actual $Y_X \in N$; they do not automatically supply elements of N.

5 An explicit conditional implementing-unitary budget

We now join the finite estimates to the source's actual weak-limit interface, rather than asserting that a finite matrix proves its entire infinite-algebra theorem.

Proposition 5.1 (Conservative explicit obstruction)**.** *Assume the source tree representation, product-group construction, and specified weak limits described in Section 1. Use the clock and shift on $\mathbb{C}^n$, $n \geq 3$, and let $C_d = 40768/3$. If $uMu^* \subseteq N$, then, with $\nu_n = 2\sin(\pi/n)$,*

$$C_d\nu_n + 2\|u - I\| > 1/500. \tag{19}$$

Consequently $C_d\nu_n \leq 1/1000$ implies $\|u - I\| > 1/2000$. The sufficient size $n \geq \lceil 2000\pi C_d \rceil$ is conservative and not an efficiency claim.

Proof. Write $Z(g) = W_f(\lambda(g) \otimes I)W_f^*$, $\varepsilon = \|u - I\|$, and $\beta = C_d\nu_n + 2\varepsilon$. By (16), including its group-element specialization,

$$\|uZ(g)u^* - \lambda(g) \otimes f(g)\| \leq \beta.$$

Because $uZ(g)u^* \in N$, the formula

$$\Sigma(g) = uZ(g)u^*(\rho(g) \otimes I)$$

is a representation of G. Its two factor von Neumann algebras $\mathcal{A}_1, \mathcal{A}_2$ commute, and

$$\|\Sigma(g) - \lambda(g)\rho(g) \otimes f(g)\| \leq \beta.$$

For a single factor generator s_j, conjugation $\lambda(s_j)\rho(s_j)$ on $\ell^2(F)$ converges weakly to p_e: its identity vector is fixed, and each other pair of fixed basis vectors is eventually separated. The source's separate weak limits for a, b, together with this tensor factorization, give comparison limits $tE_i\widetilde{R}_i$, $t = 1/2$. Here the commuting projections E_i have common corner $D = \mathbb{C}^n$, and the restrictions of $\widetilde{R}_1, \widetilde{R}_2$ are U_n, V_n.

Take weak-operator cluster points B_i of the bounded sequences $\Sigma(s_j, e)$ and $\Sigma(e, s_j)$ separately. They lie in $\mathcal{A}_i$, and weak closedness of norm balls gives $\left\|B_i - tE_i\widetilde{R}_i\right\| \leq \beta$. This uses neither convergence of products of arbitrary weakly convergent operators nor a commuting limit of the original a, b. Put $T_i = 2B_i$. If $\beta \leq 1/500$, their errors are at most $1/250$, contradicting Corollary 3.3. The final implication follows from $\sin(\pi/n) \leq \pi/n$. □

The source construction already supplies the abstract embedding by W_f^*. The proposition only excludes a specified small neighborhood of the identity. Its proof uses the exact infinite representation and weak-limit hypotheses; a finite collection of sampled group elements or finite-dimensional compressions does not establish those hypotheses. The constants could be improved by using the exact $j(e)$ and dimension-dependent $\cos(\pi/n)$ in place of the convenient uniform threshold. We have not optimized that multi-stage nonlinear budget.

6 Physical norms, examples, and failure controls

6.1 The projection bound is attained

Let $0 < c \leq 1$, $s \geq 0$, $c^2 + s^2 = 1$, and

$$E = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad P = \begin{pmatrix} c^2 & cs \\ cs & s^2 \end{pmatrix}.$$

Then

$$S = PE + (I-P)(I-E) = c\begin{pmatrix} c & -s \\ s & c \end{pmatrix}, \quad J = \begin{pmatrix} c & -s \\ s & c \end{pmatrix}.$$

Thus $\|P-E\| = s$ and $\|J-I\| = \sqrt{2-2c} = j(s)$. Rational parametrization $c = (1-v^2)/(1+v^2)$, $s = 2v/(1+v^2)$, $0 \le v < 1$, supplies exact fixtures without floating-point spectral decisions. At $v = 1$, the projections are orthogonal and $S = 0$; the strict $e < 1$ condition is genuine. For $n = 4$, the clock separation is $1/\sqrt{2}$, whereas the identity-centered branch argument alone gives $1 - \nu_4/2 = 1 - 1/\sqrt{2}$. This compares two sufficient path arguments, not the true distance to the commuting-pair set.

6.2 Why arbitrary finite compression cannot replace the corner

Commuting self-adjoint matrices can have noncommuting compressions. Take $A = \mathrm{diag}(1,0,0)$, $B = \mathrm{diag}(0,1,0)$, and $P = I - \mathbf{1}\mathbf{1}^*/3$, where $\mathbf{1} = (1,1,1)^T$. Writing $a = Pe_1, b = Pe_2$, we have $\langle a, b\rangle = -1/3$ and

$$[PAP, PBP] = -\tfrac{1}{3}(ab^* - ba^*) \neq 0.$$

The source corner proof instead uses the special spectral projections $Q_i \in \mathcal{A}_i$ and the mutual commutation of $\mathcal{A}_1, \mathcal{A}_2$. Deleting these hypotheses invalidates the extraction even when all computed Gram matrices are positive.

6.3 Nonorthonormal coordinates and omitted residuals

Here the original physical-metric chapter [6], not a later comparative report, provides the relevant finite-coordinate rule. Let $J_0 : \mathbb{C}^q \to H$ be injective with Gram matrix $G_0 = J_0^* J_0 > 0$. On its invariant range an operator has coefficient matrix A_c; its physical norm is

$$\|A_c\|_{G_0} = \left\| G_0^{1/2} A_c G_0^{-1/2} \right\|_2 .$$

A physical unitary obeys $A_c^* G_0 A_c = G_0$, not necessarily $A_c^* A_c = I$. Similarity preserves spectrum and trace, but not Euclidean operator norm. If $aG_0 \preceq \widehat{G} \preceq bG_0$, $0 < a \le b$, then

$$\|A_c\|_{G_0} \le \sqrt{b/a}\, \|A_c\|_{\widehat{G}} .$$

This follows by bounding the output and input quadratic forms separately. Thus a branch or corner certificate in approximate coordinates must include this factor, and must separately certify the relevant unitarity and invariance premises.

For a bounded physical operator A, let $P_0 = J_0 G_0^{-1} J_0^*$ and $B_c = G_0^{-1} J_0^* A J_0$. Orthogonality gives the exact residual identity

$$\|AJ_0 c - J_0 \lambda c\|^2 = \|(B_c - \lambda I)c\|_{G_0}^2 + \|(I - P_0)AJ_0 c\|^2 . \tag{20}$$

The second term cannot be inferred from finite Gram positivity. This is a classical orthogonal decomposition, included as a boundary on what a finite calculation can certify, not credited as a new contribution.

7 Comparison, current context, and research limits

The branch obstruction is classical. Exel's trace-log/winding treatment [2], including its Lemma 3.1, uses the opposite commutator ordering $vuv^{-1}u^{-1}$. We use UVU^*V^*, making the source clock

invariant +1. Changing ordering changes the sign convention, not the obstruction. Our finite refinement centers the perturbation at ωI, retains its actual distance from the cut, and combines it with explicit corner losses. It does not introduce a new Bott invariant.

The primary literature also has substantial positive results under structural hypotheses. The theorem statements and abstract in [3] were inspected, including the small-near-inclusion result for amenable algebras; their full external proofs were not audited. The current arXiv record for [4] lists version 3, February 26, 2014, and the subsequent Duke publication. Its abstract concerns particular nonamenable factors stable under sufficiently small perturbations, not unrestricted one-sided near inclusions. These results are not contradicted merely by the headline of the assigned source. The assigned source has the face date October 5, 2026 and is a cached publisher preprint; no independent publication or acceptance claim is made here. Parent-supplied captured-tree byte identity is provenance, not a live-branch or proof-validity certificate.

The freshly read original continuation argument [5] distinguishes a single connected branch domain and dominated or locally uniform parameter dependence from unsupported continuation assertions. In the present problem the analogous concrete requirements are a cut-free commutator path and absolutely summable operator-valued Fourier series. Holomorphic uniqueness does not supply stability from noisy samples. The explicit gap, logarithm remainder, and Fourier tails above are the additional quantitative information; they are proved directly.

Advantages and costs. The capped Fourier estimate improves the source majorant before any truncation, and preserves matrix-level uniformity. The corner certificate identifies which spectral, intertwiner, and polar margins a computation must retain. The branch estimate permits dimension-aware certified rounding of an unnormalized trace. Their costs are substantial: the explicit n threshold is large, norm enclosures must be physical, and whole-algebra and weak-limit obligations remain infinite-dimensional. No timing, optimality, or computational speedup has been measured.

Extensions with missing inputs. A sharper envelope for the actual Fourier coefficients could reduce C_d further, but would require a uniform proof over every group word. A better corner budget might use the joint spectral geometry of T_i instead of successive triangle inequalities. It would still require commuting source algebras, not just almost commuting finite compressions. General commutators may use their exact branch gap rather than a scalar center, provided a full norm or resolvent certificate controls the path. None of these improvements by itself extends an amenability theorem, proves two-sided instability, or makes a near-identity linear map spatial.

8 Evidence and contribution boundary

All seven cached public texts were literally displayed and read: `main.tex` (54 lines), introduction (117), circle (132), near (159), obstruction (257), references (119), and README (18), totaling 856 lines. No public-source text is omitted from this assigned inventory. The fresh confidential original reads are [5], lines 1155–1244 of 2244, and [6], lines 1–210 of 210, totaling 300 lines. The 2154 complementary lines of the former, linked figures, other private volumes, and external cited proofs not expressly identified above are unread for this report. File discovery and prior report reading are not counted as proof reading.

The owned executable checks separately record exact rational algebra and floating-point diagnostics: 28 exact cases with 154 assertions, and 13 numerical cases with 38 assertions. They test majorant sums and tails, projection identities, the conservative obstruction budget, physical-coordinate identities, and finite clock examples. They do not establish uniform group-word bounds,

weak cluster points, or the source's global construction. Precise counts and hashes appear in the final owned receipts. The local increment is an explicit collection of linked quantitative budgets, including a capped source majorant and its amplified tail control. This merits the conservative C2 assessment, separately from the source's L3 applicability. The infinite representation, weak-limit and embedding architecture remain source inputs, even where their algebraic interfaces have been rederived here.

References

[1] OpenAI mathematical preprint collection, *Near Inclusions of von Neumann Algebras Without Small Spatial Embeddings*, October 5, 2026. Assigned cached manuscript, seven texts; publisher repository `https://github.com/openai/math`.

[2] R. Exel, *The soft torus and applications to almost commuting matrices*, Pacific J. Math. 160 (1993), 207–217. `https://msp.org/pjm/1993/160-2/pjm-v160-n2-p02-p.pdf`. Lemma 3.1 and its proof inspected; not a full-paper audit.

[3] J. Cameron, E. Christensen, A. M. Sinclair, R. R. Smith, S. White, and A. D. Wiggins, *Type II_1 factors satisfying the spatial isomorphism conjecture*, Proc. Natl. Acad. Sci. USA 109 (2012), 20338–20343. Author manuscript `https://eprints.gla.ac.uk/72738/1/id72738.pdf`.

[4] J. Cameron, E. Christensen, A. M. Sinclair, R. R. Smith, S. White, and A. D. Wiggins, *Kadison–Kastler stable factors*, Duke Math. J. 163 (2014), 2639–2686. `https://arxiv.org/abs/1209.4116`. Current metadata and abstract inspected.

[5] M. Subbarao, original private manuscript `TR-rev-1-old-tr018.tex`, RamFoundationTRs, literal lines 1155–1244: branch-domain continuation and its limitations. Local confidential source; not uploaded.

[6] M. Subbarao, original private chapter *Physical metrics*, RamCalBooks, volume 21, `ch01d-physical-metrics.tex`, literal lines 1–210. Local confidential source; not uploaded.